Project 9 : full Incident Simulation

1. Objective

The objective of this project is to simulate and investigate a cybersecurity incident using SOC monitoring and digital forensic techniques. The investigation includes log analysis, network traffic analysis, disk forensics, memory analysis, and incident reporting.

2. Tools Used

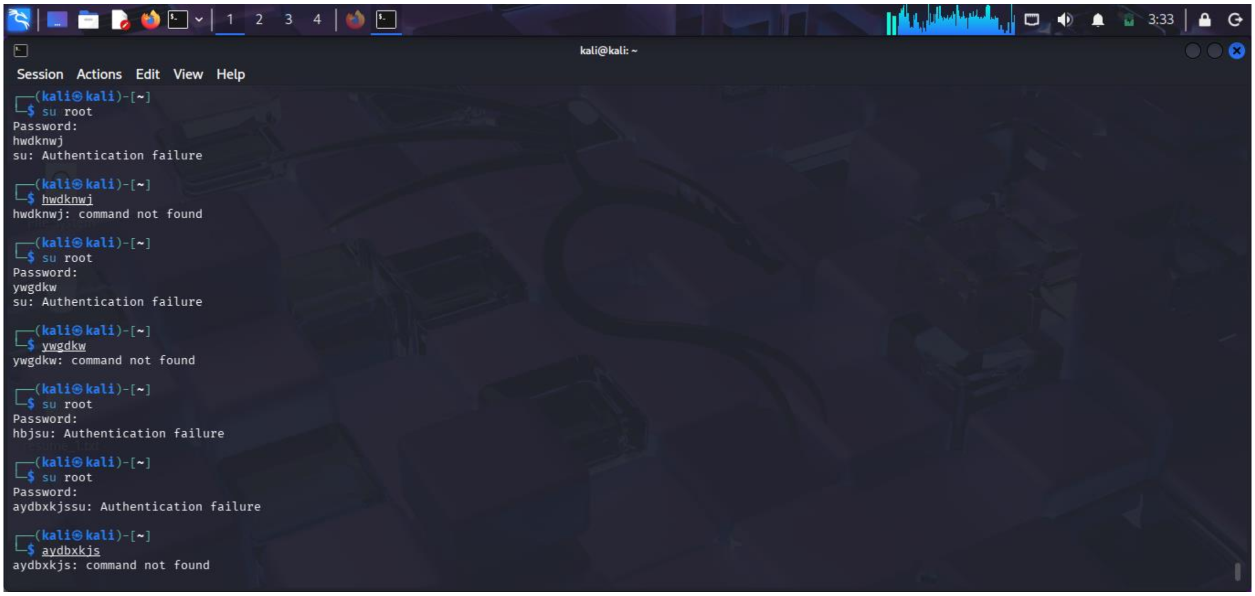
Tool	Purpose
Kali Linux	Investigation environment
VirtualBox	Virtual lab setup
Wireshark	Network traffic analysis
Autopsy	Disk forensic investigation
Volatility Framework	Memory forensic analysis
Linux Log Utilities	Log analysis

3. Incident Scenario

A simulated attack scenario was created in the lab environment. The attacker performed multiple failed login attempts and suspicious activities were identified through system logs and forensic investigation.

Activities performed:

- Failed SSH login attempts
- Suspicious network activity
- File deletion simulation
- Memory artifact analysis



- Screenshot of lab setup or terminal activity

4. Log Analysis Investigation

Authentication logs were analyzed using Linux log utilities to identify suspicious login attempts.

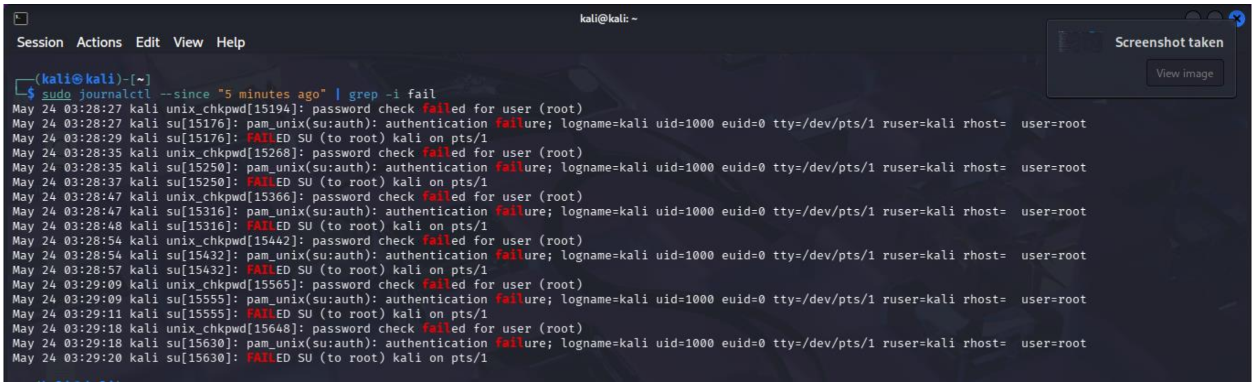
Commands Used

```
grep "Failed password" /var/log/auth.log
cat /var/log/auth.log | grep "Failed"
```

Findings

- Multiple failed login attempts were identified.
- Repeated authentication failures indicate possible brute-force activity.

- Log entries contained timestamps and usernames used during login attempts.



- Failed password log output
- Terminal screenshot showing grep results

5. Network Traffic Analysis

Network traffic analysis was performed using Wireshark.

Procedure

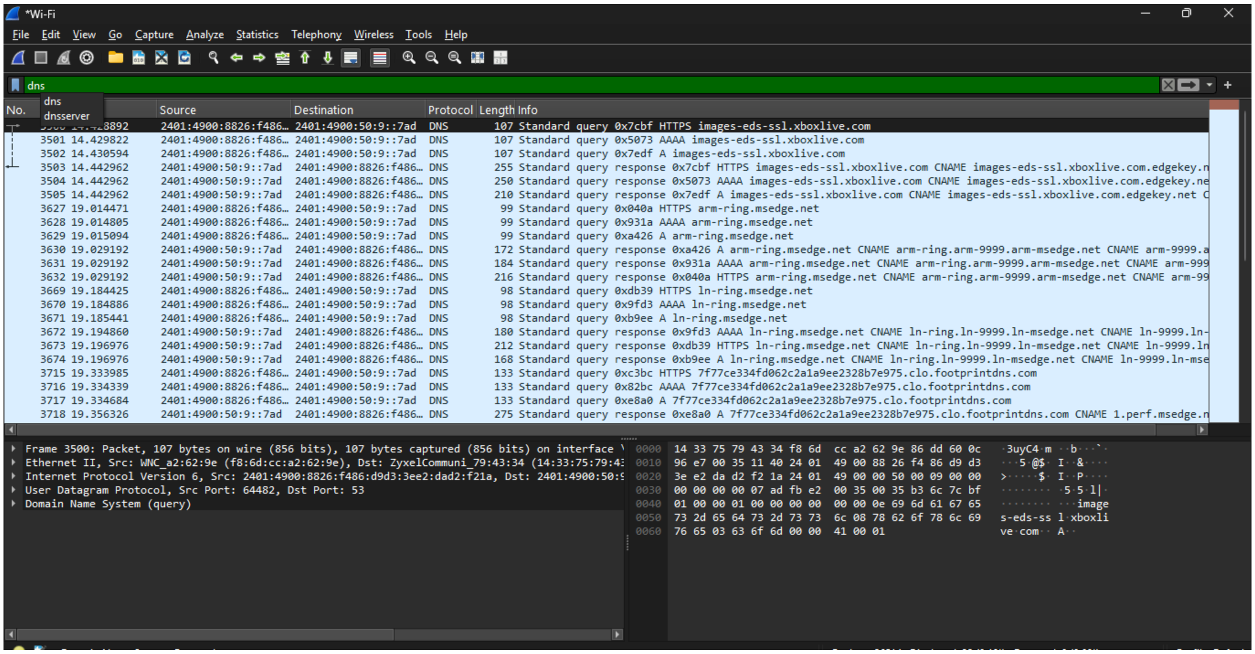
- Packet capture was started on the active network adapter.
- ICMP traffic was generated using ping commands.
- DNS packets were analyzed using Wireshark filters.

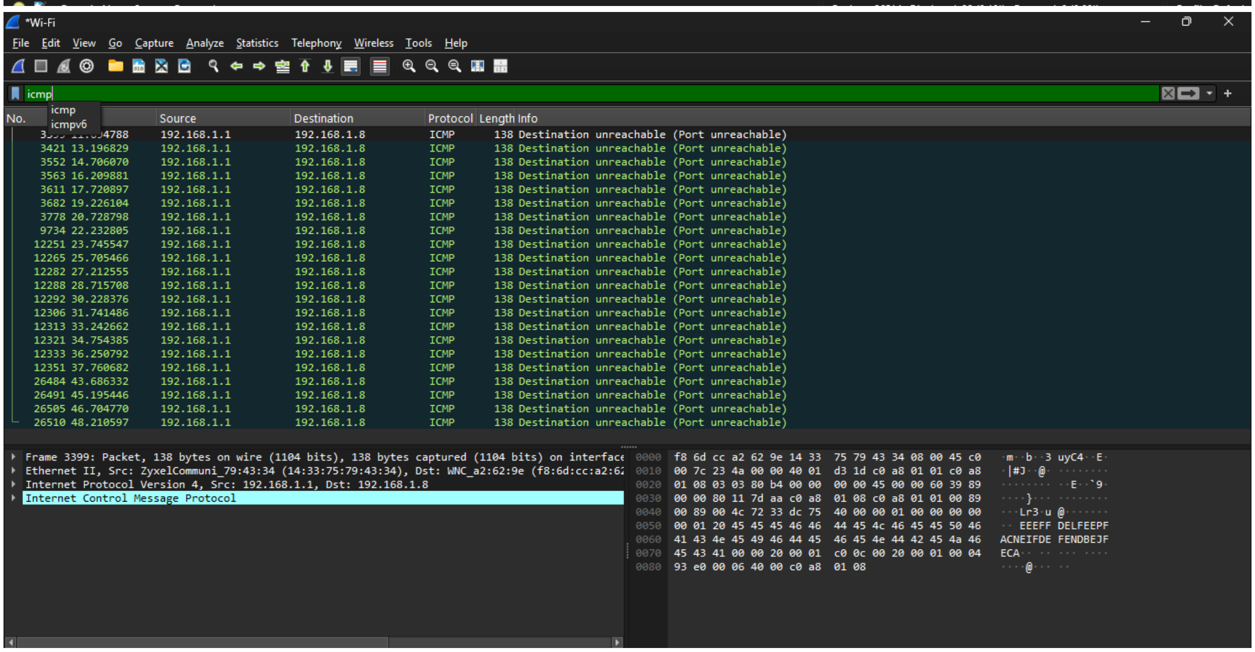
Filters Used

```
icmp
dns
```

Findings

- ICMP packets confirmed active communication.
- DNS queries and responses were successfully captured.
- Traffic analysis demonstrated packet-level investigation capability.





- Wireshark live capture
- ICMP filter results
- DNS filter results

6. Disk Forensics Investigation

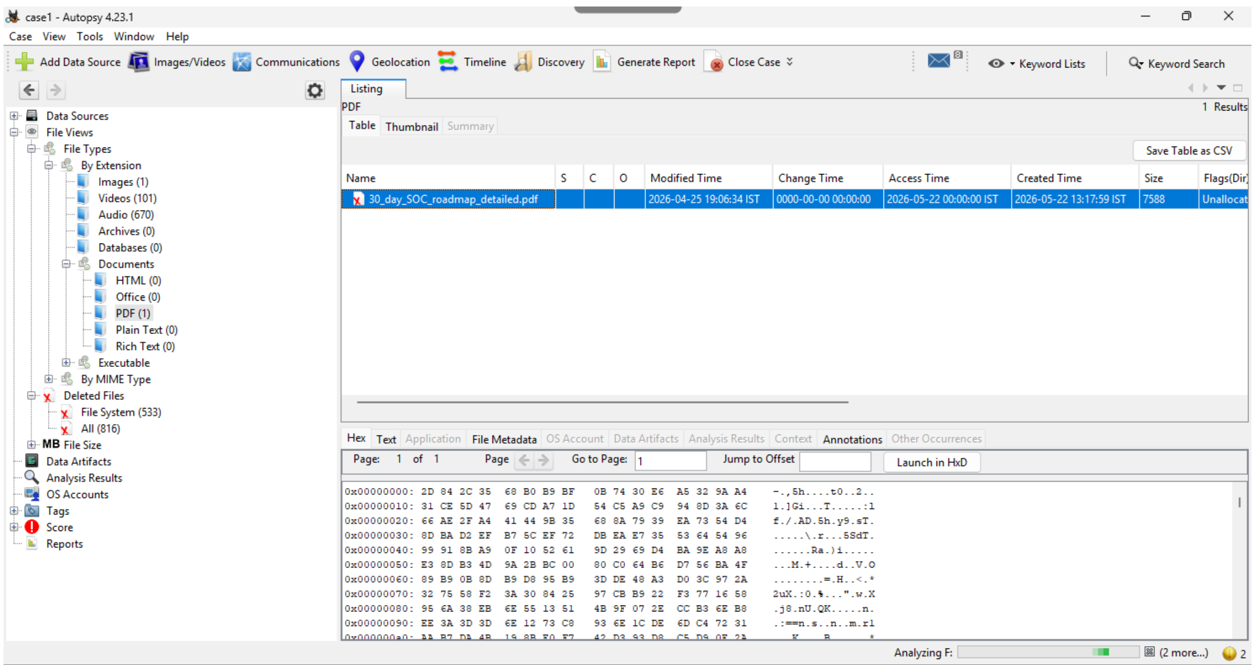
Disk forensic analysis was performed using Autopsy.

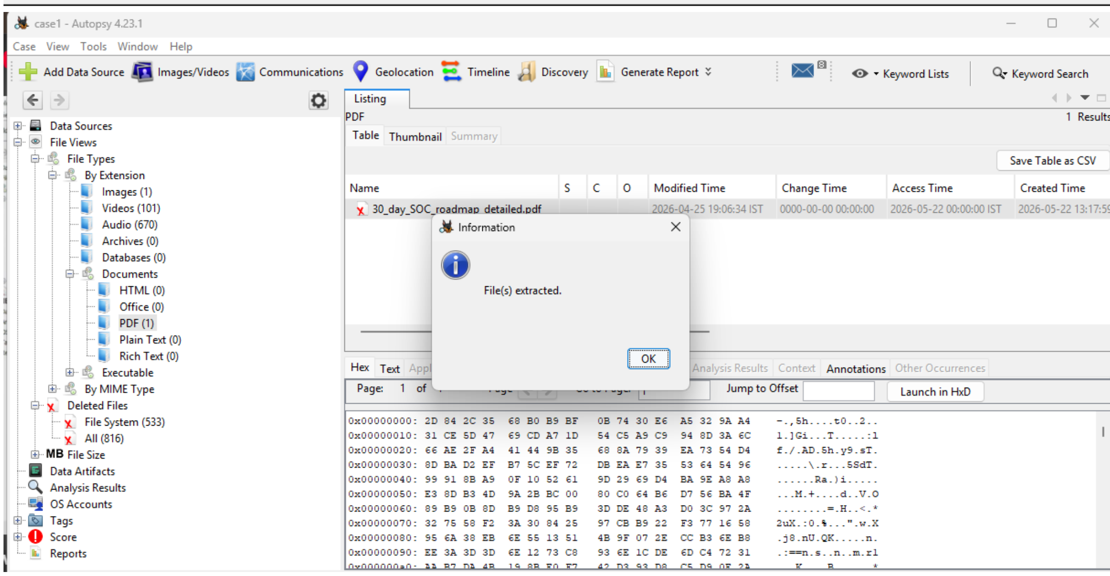
Investigation Performed

- Deleted file analysis
- Metadata analysis
- Timeline analysis
- File activity investigation

Findings

- Deleted files were identified successfully.
- File timestamps and metadata were recovered.
- Timeline analysis showed user activity patterns.





- Autopsy case creation
- Deleted files section
- Metadata analysis
- Timeline analysis

7. Memory Forensics Investigation

Memory analysis was conducted using the Volatility Framework.

Commands Used

```
vol.py -f memory.raw windows.pslist
vol.py -f memory.raw windows.netscan
vol.py -f memory.raw windows.cmdline
```

Findings

- Running processes were identified.
- Active network connections were analyzed.
- Memory artifacts related to system activity were recovered.

Windows PowerShell

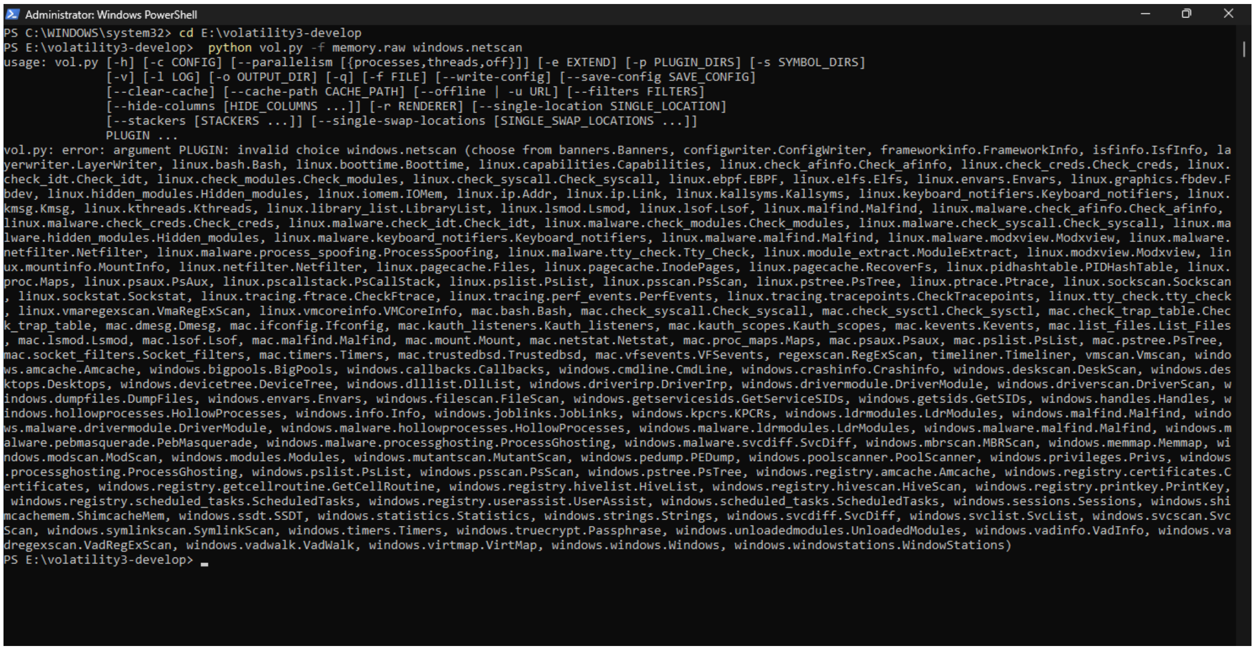
```
PS E:\volatility3-develop> python vol.py -f memory.raw windows.pslist
Volatility 3 Framework 2.28.1
Progress:  100.00               PDB scanning finished
PID     PPID    ImageFileName   Offset(V)       Threads Handles SessionId       Wow64   CreateTime      ExitTime        File output

4       0       System  0xa98817585040  286     -       N/A     False   2026-05-11 04:03:37.000000 UTC  N/A     Disabled
140     4       Secure System   0xa988175df080  0       -       N/A     False   2026-05-11 04:03:35.000000 UTC  N/A     Disabled
184     4       Registry        0xa98817604080  4       -       N/A     False   2026-05-11 04:03:35.000000 UTC  N/A     Disabled
624     4       smss.exe        0xa9881c94f040  2       -       N/A     False   2026-05-11 04:03:38.000000 UTC  N/A     Disabled
944     816     csrss.exe       0xa988276d6140  14      -       0       False   2026-05-11 04:03:44.000000 UTC  N/A     Disabled
924     816     wininit.exe     0xa9882e276180  2       -       0       False   2026-05-11 04:03:45.000000 UTC  N/A     Disabled
820     916     csrss.exe       0xa9882e28b180  0       -       1       False   2026-05-11 04:03:45.000000 UTC  2026-05-11 08:06:47.000000 UTC  Disabled
1140    924     services.exe    0xa9882e298180  8       -       0       False   2026-05-11 04:03:46.000000 UTC  N/A     Disabled
1176    924     LsaIso.exe      0xa98827638280  1       -       0       False   2026-05-11 04:03:46.000000 UTC  N/A     Disabled
1188    924     lsass.exe       0xa9882c383100  12      -       0       False   2026-05-11 04:03:46.000000 UTC  N/A     Disabled
1316    1140    svchost.exe     0xa9882e1480c0  29      -       0       False   2026-05-11 04:03:47.000000 UTC  N/A     Disabled
1352    924     fontdrvhost.ex  0xa9882e1660c0  5       -       0       False   2026-05-11 04:03:47.000000 UTC  N/A     Disabled
1452    1140    svchost.exe     0xa9882e1570c0  13      -       0       False   2026-05-11 04:03:47.000000 UTC  N/A     Disabled
1508    1140    svchost.exe     0xa9881fc460c0  6       -       0       False   2026-05-11 04:03:47.000000 UTC  N/A     Disabled
1692    1140    svchost.exe     0xa9881fe9b0c0  2       -       0       False   2026-05-11 04:03:47.000000 UTC  N/A     Disabled
1736    1140    svchost.exe     0xa988250240c0  5       -       0       False   2026-05-11 04:03:47.000000 UTC  N/A     Disabled
1832    1140    svchost.exe     0xa988251020c0  11      -       0       False   2026-05-11 04:03:47.000000 UTC  N/A     Disabled
1840    1140    svchost.exe     0xa988250bd0c0  3       -       0       False   2026-05-11 04:03:47.000000 UTC  N/A     Disabled
1844    1140    svchost.exe     0xa988250ce0c0  6       -       0       False   2026-05-11 04:03:47.000000 UTC  N/A     Disabled
1860    1140    svchost.exe     0xa988252ac0c0  11      -       0       False   2026-05-11 04:03:47.000000 UTC  N/A     Disabled
1912    1140    svchost.exe     0xa988251bd0c0  4       -       0       False   2026-05-11 04:03:47.000000 UTC  N/A     Disabled
1948    1140    svchost.exe     0xa988253680c0  8       -       0       False   2026-05-11 04:03:47.000000 UTC  N/A     Disabled
1996    1140    svchost.exe     0xa988254ce0c0  5       -       0       False   2026-05-11 04:03:47.000000 UTC  N/A     Disabled
2020    1140    svchost.exe     0xa9882548a0c0  10      -       0       False   2026-05-11 04:03:47.000000 UTC  N/A     Disabled
2068    1140    svchost.exe     0xa988258240c0  19      -       0       False   2026-05-11 04:03:47.000000 UTC  N/A     Disabled
2088    1140    svchost.exe     0xa988257bd0c0  5       -       0       False   2026-05-11 04:03:47.000000 UTC  N/A     Disabled
2112    1140    svchost.exe     0xa988257790c0  11      -       0       False   2026-05-11 04:03:47.000000 UTC  N/A     Disabled
2360    1140    svchost.exe     0xa98825a790c0  11      -       0       False   2026-05-11 04:03:47.000000 UTC  N/A     Disabled
2544    1140    svchost.exe     0xa988260020c0  6       -       0       False   2026-05-11 04:03:47.000000 UTC  N/A     Disabled
2772    1140    svchost.exe     0xa988259df0c0  6       -       0       False   2026-05-11 04:03:47.000000 UTC  N/A     Disabled
2848    1140    atiesrxx.exe    0xa988265020c0  26      -       0       False   2026-05-11 04:03:48.000000 UTC  N/A     Disabled
2856    1140    amdfendrsr.exe  0xa988265130c0  4       -       0       False   2026-05-11 04:03:48.000000 UTC  N/A     Disabled
2864    1140    svchost.exe     0xa988264bd0c0  9       -       0       False   2026-05-11 04:03:48.000000 UTC  N/A     Disabled
2892    1140    svchost.exe     0xa988264460c0  3       -       0       False   2026-05-11 04:03:48.000000 UTC  N/A     Disabled
3056    1140    svchost.exe     0xa988268570c0  6       -       0       False   2026-05-11 04:03:48.000000 UTC  N/A     Disabled
3064    1140    svchost.exe     0xa988268ac0c0  5       -       0       False   2026-05-11 04:03:48.000000 UTC  N/A     Disabled
1812    1140    svchost.exe     0xa988268350c0  4       -       0       False   2026-05-11 04:03:48.000000 UTC  N/A     Disabled
3096    1140    svchost.exe     0xa988268df0c0  3       -       0       False   2026-05-11 04:03:48.000000 UTC  N/A     Disabled
3108    4       MemCompression  0xa98827694280  66      -       N/A     False   2026-05-11 04:03:48.000000 UTC  N/A     Disabled
3168    1140    svchost.exe     0xa98826a020c0  4       -       0       False   2026-05-11 04:03:48.000000 UTC  N/A     Disabled
3176    1140    svchost.exe     0xa988269df0c0  9       -       0       False   2026-05-11 04:03:48.000000 UTC  N/A     Disabled
3264    1140    svchost.exe     0xa98826d9b0c0  11      -       0       False   2026-05-11 04:03:48.000000 UTC  N/A     Disabled
3312    1140    svchost.exe     0xa98826cdf0c0  7       -       0       False   2026-05-11 04:03:48.000000 UTC  N/A     Disabled
```

Windows PowerShell

```
PS E:\volatility3-develop> python vol.py -f memory.raw windows.cmdline
Volatility 3 Framework 2.28.1
Progress:  100.00               PDB scanning finished
PID     Process Args

4       System  -
140     Secure System   -
184     Registry        -
624     smss.exe        \SystemRoot\System32\smss.exe
944     csrss.exe       %SystemRoot%\system32\csrss.exe ObjectDirectory=\Windows SharedSection=1024,20480,768 Windows=On SubSystemType=Windows ServerDll=basesrv,1 ServerDll=winsrv:UserServer
DllInitialization,3 ServerDll=sxssrv,4 ProfileControl=Off MaxRequestThreads=16
924     wininit.exe     -
820     csrss.exe       -
1140    services.exe    C:\WINDOWS\system32\services.exe
1176    LsaIso.exe      -
1188    lsass.exe       C:\WINDOWS\system32\lsass.exe
1316    svchost.exe     C:\WINDOWS\system32\svchost.exe -k DcomLaunch -p
1352    fontdrvhost.ex  -
1452    svchost.exe     C:\WINDOWS\system32\svchost.exe -k RPCSS -p
1508    svchost.exe     C:\WINDOWS\system32\svchost.exe -k DcomLaunch -p -s LSM
1692    svchost.exe     C:\WINDOWS\system32\svchost.exe -k LocalSystemNetworkRestricted -p -s HvHost
1736    svchost.exe     C:\WINDOWS\system32\svchost.exe -k LocalService -p -s nsi
1832    svchost.exe     C:\WINDOWS\system32\svchost.exe -k NetworkService -p
1840    svchost.exe     C:\WINDOWS\system32\svchost.exe -k LocalServiceNetworkRestricted -p -s TimeBrokerSvc
1844    svchost.exe     C:\WINDOWS\System32\svchost.exe -k LocalSystemNetworkRestricted -p -s NcbService
1860    svchost.exe     C:\WINDOWS\system32\svchost.exe -k netsvcs -p -s Schedule
1912    svchost.exe     C:\WINDOWS\system32\svchost.exe -k LocalServiceNoNetwork -p
1948    svchost.exe     C:\WINDOWS\system32\svchost.exe -k LocalServiceNetworkRestricted -p -s Dhcp
1996    svchost.exe     C:\WINDOWS\system32\svchost.exe -k UserProfileService -p -s ProfSvc
2020    svchost.exe     C:\WINDOWS\system32\svchost.exe -k LocalSystemNetworkRestricted -p -s DisplayEnhancementService
2068    svchost.exe     C:\WINDOWS\System32\svchost.exe -k netprofm -p -s netprofm
2088    svchost.exe     C:\WINDOWS\system32\svchost.exe -k LocalServiceHttp -p
2112    svchost.exe     -
2360    svchost.exe     C:\WINDOWS\system32\svchost.exe -k LocalServiceNetworkRestricted -p -s Wcmsvc
2544    svchost.exe     C:\WINDOWS\system32\svchost.exe -k appmodel -p -s StateRepository
2772    svchost.exe     C:\WINDOWS\system32\svchost.exe -k NetSvcs -p -s hns
2848    atiesrxx.exe    -
2856    amdfendrsr.exe  -
2864    svchost.exe     C:\WINDOWS\System32\svchost.exe -k LocalServiceNetworkRestricted -p -s EventLog
2892    svchost.exe     C:\WINDOWS\system32\svchost.exe -k LocalService -p -s DispBrokerDesktopSvc
3056    svchost.exe     C:\WINDOWS\system32\svchost.exe -k LocalServiceRedirectionGuard -p -s EventSystem
3064    svchost.exe     C:\WINDOWS\system32\svchost.exe -k LocalSystemNetworkRestricted -p -s SysMain
1812    svchost.exe     C:\WINDOWS\System32\svchost.exe -k netsvcs -p -s Themes
3096    svchost.exe     C:\WINDOWS\system32\svchost.exe -k netsvcsRedirectionGuard -p -s SENS
3108    MemCompression  -
3168    svchost.exe     C:\WINDOWS\System32\svchost.exe -k LocalSystemNetworkRestricted -p -s AudioEndpointBuilder
3176    svchost.exe     C:\WINDOWS\system32\svchost.exe -k LocalService -p -s FontCache
3264    svchost.exe     C:\WINDOWS\System32\svchost.exe -k LocalServiceNetworkRestricted -p
```



- Volatility command execution
- Process list output
- Network scan results

8. Evidence Collected

The following evidence was collected during the investigation:

- Authentication logs
- Wireshark packet captures
- Disk forensic screenshots
- Memory analysis outputs
- Timeline evidence

9. Incident Response Actions

The following response actions were considered:

- Investigation of failed login attempts

- Analysis of suspicious network activity
- Identification of deleted files
- Examination of memory artifacts
- Documentation of evidence

10. Conclusion

The simulated incident was successfully investigated using SOC monitoring and digital forensic techniques. Logs, network traffic, disk evidence, and memory artifacts were analyzed to identify suspicious activity. The project demonstrated practical skills in cybersecurity investigation and incident response.